

COMPUTER NETWORK PROTOCOLS

COMPLETE STUDY NOTES

For:

**Government CBT / MCQ Exam
Information Security Scientist B — Computer Science**

Covers: OSI Layers 2–7 | Application | Security | Transport | Network | Routing | Data Link

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SECTION 1: APPLICATION LAYER PROTOCOLS



⚠ **EXAM TIP:** Application layer = Layer 7 of OSI model. Most MCQ questions on protocols come from this layer!



Port & Protocol Quick Reference Table

Protocol	Port(s)	Layer	Secure?	Key Feature
HTTP	80	Application	NO ⚠	Stateless, request-response
HTTPS	443	Application	YES ✓	HTTP + TLS encryption
FTP	20 (data), 21 (ctrl)	Application	NO ⚠	File transfer
SFTP	22	Application	YES ✓	FTP over SSH
SMTP	25 / 587	Application	Partial	Send email
POP3	110	Application	Basic	Download & delete from server
IMAP	143	Application	Better	Sync email, stays on server
DNS	53	Application	Vuln	Name → IP mapping
Telnet	23	Application	NO ⚠	Remote login, NO encryption
SSH	22	Application	YES ✓	Secure remote login
SNMP	161/162	Application	Basic	Device monitoring

req-res

file transfer

email

remote login

● HTTP / HTTPS

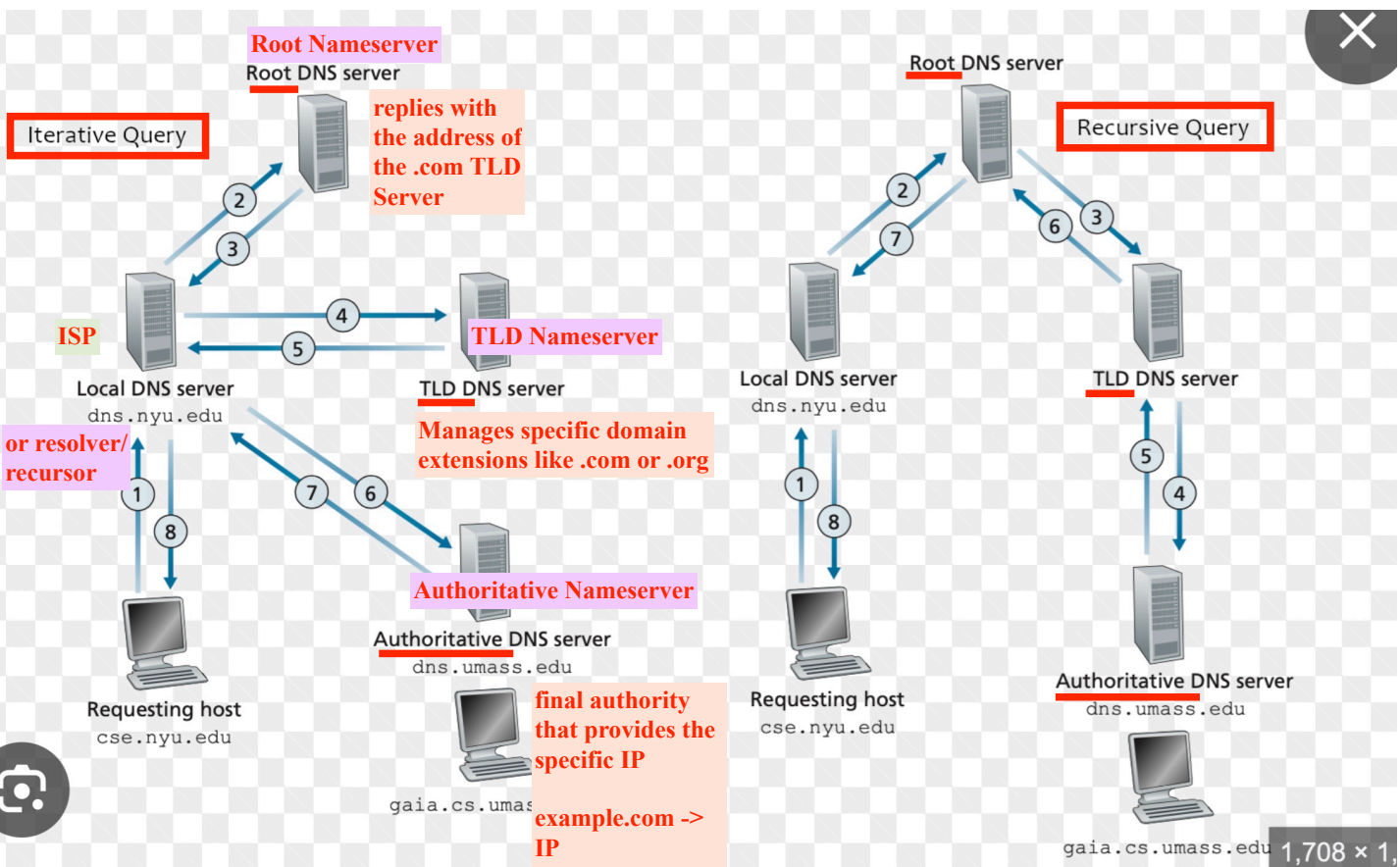
- ▶ **Full Form:** HyperText Transfer Protocol / HTTP Secure
- ▶ **HTTP Port:** 80
- ▶ **HTTPS Port:** 443
- ▶ **Nature:** Stateless — each request is independent (no memory of previous requests)
- ▶ **Model:** Request-Response (Client sends request → Server sends response)
- ▶ **HTTPS =:** HTTP + TLS (Transport Layer Security) for encryption

✓ **KEY POINT:** HTTPS uses TLS (not SSL — though often called SSL loosely)

⚠ **EXAM TIP:** HTTP is INSECURE. HTTPS is SECURE. Port 80 vs 443 — common MCQ!

● DNS — Domain Name System

- ▶ **Function:** Translates domain names → IP addresses (e.g., google.com → 142.250.x.x)
- ▶ **Port:** 53 (both UDP and TCP)
- ▶ **Protocol used:** UDP for queries, TCP for zone transfers
 - DNS Hierarchy: Root → TLD (.com, .in) → Authoritative → Recursive Resolver



⚠ DNS Attacks (CRITICAL for Security Exam)

Attack	What Happens	Impact
DNS Spoofing	Fake DNS responses sent to client	Redirected to malicious site
DNS Cache Poisoning	Corrupt DNS cache with wrong IP	Persistent redirection to attacker
DNS Amplification	Small query → large response (DDoS)	Bandwidth exhaustion
DNS Hijacking	Attacker controls DNS server	All users redirected

🟡 FTP — File Transfer Protocol

- ▶ **Ports:** Port 21 = Control/Commands | Port 20 = Data transfer
- ▶ **Security:** NO encryption — credentials sent in plaintext
- ▶ **Modes:** **Active** (server connects to client) | **Passive** (client connects to server)
 - **SFTP** = FTP over SSH (Port 22) — SECURE version
 - **FTPS** = FTP over SSL/TLS — another secure variant

⚠ **EXAM TIP: FTP uses TWO ports (20 & 21). Very common MCQ question!**

✉ Email Protocols — SMTP, POP3, IMAP

Protocol	Port	Direction	Stores on Server?	Use Case
SMTP	25/587	OUTGOING (Send)	No	Sending emails
POP3	110	INCOMING (Receive)	NO — downloads & deletes	Single device use
IMAP	143	INCOMING (Receive)	YES — stays on server	Multi-device sync

📧 Email Flow (Sender → Recipient)

- [1] → Sender composes email in **MUA** (Mail User Agent)
- [2] → MUA sends via **SMTP** to Sender's Mail Server
- [3] → Sender's Mail Server uses **SMTP** to deliver to Recipient's Mail Server
- [4] → Recipient retrieves email using POP3 or IMAP

⚠ **EXAM TIP: SMTP = Sending. POP3/IMAP = Receiving. This distinction is heavily tested!**

🔴 Telnet vs SSH — Security Comparison

Feature	Telnet	SSH
Port	23	22
Encryption	NONE — plaintext ⚠	Full encryption ✓
Authentication	Basic	Strong (keys/passwords)

Feature	Telnet	SSH
Security	INSECURE — VULNERABLE	SECURE ✓
Usage	Legacy/deprecated	Current standard
MCQ Note	Classic 'what is wrong with Telnet?' answer	Replacement for Telnet

⚠ **EXAM TIP: Telnet's biggest flaw = NO ENCRYPTION. SSH replaces Telnet. Port 22 vs 23!**

● SNMP — Simple Network Management Protocol

- ▶ **Purpose:** Monitor and manage network devices (routers, switches, servers)
- ▶ **Ports:** 161 (agent/queries) | 162 (trap messages)
- ▶ **Versions:** SNMPv1/v2 = weak security | **SNMPv3** = encryption + authentication
 - Components: Manager, Agent, **MIB (Management Information Base)**

⚠ **EXAM TIP: SNMPv3 is the secure version. Community strings in v1/v2 sent in plaintext!**



SECTION 2: SECURITY PROTOCOLS



⚠ **EXAM TIP:** This entire section is **DIRECTLY** relevant to Information Security Scientist exam. Master every protocol here!

◆ SSL / TLS — Secure Sockets Layer / Transport Layer Security

- ▶ **Purpose:** Provides encryption, authentication, and data integrity
- ▶ **OSI Layer:** Between Transport and Application (Session Layer)
- ▶ **Current Version:** TLS 1.3 (TLS has replaced SSL — SSL is deprecated!)



TLS Handshake Process

Step	Who Acts	What Happens
1. ClientHello	Client → Server	Client sends: supported cipher suites, TLS version, random number
2. ServerHello	Server → Client	Server chooses cipher suite, sends certificate
3. Certificate Verify	Client verifies	Client checks server's digital certificate with CA
4. Key Exchange	Both	Session keys generated using asymmetric crypto (RSA/ECDH)
5. Finished	Both	Symmetric encryption begins for data transfer

- ▶ **Certificates:** Issued by Certificate Authority (CA) — verifies server identity
- ▶ **Encryption:** Asymmetric for handshake → Symmetric for data (faster)
- ✓ **KEY POINT:** SSL is deprecated. TLS 1.2/1.3 are current. TLS 1.0/1.1 also deprecated.

⚠ **EXAM TIP:** SSL ≠ TLS but terms often used interchangeably in exams. Know both!

◆ IPsec — Internet Protocol Security

- ▶ **Layer:** Network Layer (Layer 3)
- ▶ **Purpose:** Secure IP communications — used in VPNs
- ▶ **Protocols:** AH (Authentication Header) + ESP (Encapsulating Security Payload)



IPsec Modes — CRITICAL COMPARISON

Mode	What is Protected?	Use Case	Header Encrypted?
Transport Mode	Payload only (data encrypted)	Host-to-Host communication	NO — IP header visible
Tunnel Mode	Entire original IP packet	VPN Gateway-to-Gateway	YES — new IP header added

Component	Function	Provides
AH (Authentication Header)	Adds authentication + integrity	No encryption, anti-replay
ESP (Encapsulating Security Payload)	Encryption + authentication + integrity	Full security
IKE (Internet Key Exchange)	Key management protocol	Negotiates security associations (SA)

⚠ **EXAM TIP: Transport Mode = host-to-host. Tunnel Mode = network-to-network (VPN). Very common MCQ!**

🔴 Kerberos — Ticket-Based Authentication

- ▶ **Purpose:** Network authentication using tickets — prevents password transmission
- ▶ **Developer:** MIT — used in Windows Active Directory
- ▶ **Port:** 88

📋 Kerberos Authentication Flow

- [1] → Client requests Ticket Granting Ticket (TGT) from KDC/AS (Authentication Server)
- [2] → KDC verifies credentials → Issues TGT (encrypted with client's key)
- [3] → Client presents TGT to TGS (Ticket Granting Server) for service ticket
- [4] → TGS issues Service Ticket for requested resource
- [5] → Client presents Service Ticket to target server → Access Granted

- ▶ **Key Components:** KDC (Key Distribution Center), AS (Authentication Server), TGS (Ticket Granting Server)

✓ **KEY POINT:** Kerberos uses SYMMETRIC encryption (shared secret keys) — NOT asymmetric

⚠ **EXAM TIP: KDC = AS + TGS. Tickets have timestamps to prevent replay attacks!**

📋 SET — Secure Electronic Transaction

- ▶ **Purpose:** Secure credit/debit card transactions over the internet
- ▶ **Developed by:** Visa + Mastercard (1990s)
- ▶ **Key Feature:** Both merchant and customer are authenticated via digital certificates
 - Merchant never sees full credit card number (uses dual signature)
 - Customer never sees merchant's account details

✓ **KEY POINT:** SET protects payment data end-to-end using dual signatures and encryption

⚠ **EXAM TIP: SET = credit card security. Uses both symmetric and asymmetric encryption.**

🔴 PGP — Pretty Good Privacy

- ▶ **Purpose:** Email encryption and digital signatures
- ▶ **Created by:** Phil Zimmermann (1991)
- ▶ **Standard:** OpenPGP (RFC 4880)

Feature	How PGP Does It
Confidentiality	Encrypt message with session key → encrypt session key with recipient's public key
Authentication	Sign message with sender's private key → recipient verifies with sender's public key
Integrity	Hash of message included in digital signature
Non-repudiation	Sender cannot deny — only their private key could sign

✓ **KEY POINT:** PGP = Web of Trust model (no central CA). GPG is the free implementation.

⚠ **EXAM TIP:** PGP uses hybrid encryption: symmetric for data, asymmetric for key exchange!



SECTION 3: TRANSPORT LAYER (Layer 4)



⚡ TCP vs UDP — Master Comparison Table

Feature	TCP	UDP
Full Name	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-ORIENTED ✓	Connectionless ✗
Reliability	RELIABLE — guarantees delivery	UNRELIABLE — best effort
Speed	Slower (overhead)	FASTER — no overhead
Ordering	In-order delivery guaranteed	No ordering guarantee
Error Check	Yes — checksum + retransmit	Checksum only (no retransmit)
Flow Control	Yes (sliding window)	No
Congestion Control	Yes	No
Header Size	20-60 bytes	8 bytes (smaller)
Use Cases	HTTP/S, FTP, SMTP, SSH	DNS, streaming, VoIP, gaming

🔵 TCP — 3-Way Handshake (SYN-SYN/ACK-ACK)

⚠️ EXAM TIP: 3-Way Handshake = most tested TCP concept. Know every step!

Step	Sender → Receiver	Flag Sent	Meaning
Step 1	Client → Server	SYN	Client: 'I want to connect, here is my sequence number'
Step 2	Server → Client	SYN + ACK	Server: 'OK, I acknowledge. Here is MY sequence number'
Step 3	Client → Server	ACK	Client: 'Acknowledged. Connection established!'

✓ KEY POINT: SYN = Synchronize | ACK = Acknowledgment | FIN = Finish (connection termination)

▶ Connection Termination: 4-Way: FIN → ACK → FIN → ACK

▶ SYN Flood Attack: Attacker sends many SYN packets without completing handshake (DoS attack)



TCP Control Mechanisms

Mechanism	Purpose	How It Works
Flow Control	Prevent sender overwhelming receiver	Receiver advertises window size (sliding window protocol)

Mechanism	Purpose	How It Works
Congestion Control	Prevent network congestion	Slow Start → Congestion Avoidance → Fast Retransmit
Error Control	Ensure data integrity	Sequence numbers + ACKs + checksum + retransmission
Multiplexing	Multiple apps share TCP	Port numbers distinguish different connections

🟡 UDP — When Speed > Reliability

► **Header Fields:** Source Port | Destination Port | Length | Checksum (only 4 fields, 8 bytes!)

UDP Use Cases:

Application	Why UDP?
DNS queries	Fast single request/response — no need for connection overhead
Video/Audio Streaming	Slight packet loss tolerable, low latency critical
VoIP (Voice over IP)	Real-time — delay worse than lost packet
Online Gaming	Speed critical, minor loss acceptable
DHCP	Broadcast-based, connection not needed
SNMP	Monitoring — lightweight queries



SECTION 4: NETWORK LAYER (Layer 3)



● IP — Internet Protocol

IPv4 vs IPv6 Comparison

Feature	IPv4	IPv6
Address Size	32-bit	128-bit
Address Format	Dotted decimal (192.168.1.1)	Hexadecimal groups (2001:db8::1)
Total Addresses	~4.3 billion (exhausted!)	~3.4 × 10 ³⁸ (virtually unlimited)
Header Size	20-60 bytes (variable)	40 bytes (fixed)
Fragmentation	By routers and hosts	By source host only
Checksum	Yes (in header)	Removed (in IPv6)
NAT Required?	YES (address shortage)	No (enough addresses)
Security	Optional (IPsec)	Mandatory (IPsec built-in)
Auto-config	DHCP	SLAAC (Stateless Address Auto-configuration)
Broadcast	Yes	No — uses Multicast instead

► **IP Fragmentation:** Large packets split into smaller fragments at router if MTU exceeded

► **MTU:** Maximum Transmission Unit — **default 1500 bytes for Ethernet**

⚠ **EXAM TIP: IPv4 = 32-bit, IPv6 = 128-bit. IPv6 has NO broadcast. Very common MCQ!**

● ICMP — Internet Control Message Protocol

► **Layer:** Network Layer (Layer 3) — works with IP

► **Purpose:** Error reporting and diagnostics (NOT for data transfer!)

ICMP Message	Type	Triggered When
Echo Request/Reply	Type 8/0	PING command — tests connectivity
Destination Unreachable	Type 3	Packet cannot reach destination
Time Exceeded	Type 11	TTL reaches 0 (Traceroute uses this!)
Redirect	Type 5	Router tells host to use better route
Source Quench	Type 4	Congestion control (deprecated)

ICMP Source Quench - ask a sender to reduce its traffic rate

✓ **KEY POINT:** PING uses ICMP Echo Request (Type 8) and Echo Reply (Type 0)

✓ **KEY POINT:** Traceroute uses TTL field + ICMP Time Exceeded messages

⚠ **EXAM TIP: ICMP Flood (Ping Flood) = DDoS attack. ICMP Smurf attack uses broadcast ping.**

🟡 ARP & RARP — Address Resolution

Protocol	Full Name	Resolves	Direction	Layer
ARP	Address Resolution Protocol	IP → MAC address	Logical → Physical	Network/Data Link boundary
RARP	Reverse ARP	MAC → IP address	Physical → Logical	Data Link
GARP	Gratuitous ARP	Announces own IP-MAC mapping	Broadcast	Used in failover/conflict detect
Proxy ARP	Proxy ARP	Router answers ARP for another host	Cross-subnet	Network

ARP Process:

- [1] → Host A needs MAC for IP 192.168.1.5 — checks ARP cache
- [2] → Not found → Broadcasts ARP Request: 'Who has 192.168.1.5?'
- [3] → Host B (192.168.1.5) replies: 'I have it! My MAC is XX:XX:XX:XX'
- [4] → Host A caches IP-to-MAC mapping in ARP table

⚠️ **EXAM TIP: ARP Poisoning/Spoofing = Security attack — attacker sends fake ARP replies to redirect traffic!**



SECTION 5: ROUTING PROTOCOLS



1 2 3 4 Routing Protocol Master Comparison

Protocol	Type	Algorithm	Metric	Scope	Admin Distance
RIP	IGP Distance Vector	Bellman-Ford	Hop Count (max 15)	Intra-domain	120
OSPF	IGP Link State	Dijkstra (SPF)	Cost (bandwidth)	Intra-domain	110
EIGRP	IGP Hybrid	DUAL	Bandwidth+Delay	Intra-domain (Cisco)	90
BGP	EGP Path Vector	Best Path Selection	AS Path + Policies	Inter-domain (Internet)	20 (eBGP)/200 (iBGP)
IS-IS	IGP Link State	Dijkstra	Cost	Intra-domain	115

● RIP — Routing Information Protocol

- ▶ **Type:** Distance Vector protocol
- ▶ **Algorithm:** Bellman-Ford
- ▶ **Metric:** Hop Count (number of routers to destination)
- ▶ **Max Hops:** 15 hops maximum — hop count of 16 = UNREACHABLE (infinity!)
- ▶ **Update Interval:** Every 30 seconds (broadcasts full routing table)
- ▶ **Versions:** RIPv1 (classful) | RIPv2 (classless, supports VLSM) | RIPv6 (IPv6)
 - RIPv1 uses **broadcast**; RIPv2 uses **multicast** (224.0.0.9)

⚠ **EXAM TIP: RIP max 15 hops = biggest limitation. NOT suitable for large networks!**

● OSPF — Open Shortest Path First ★ IMPORTANT

- ▶ **Type:** Link State protocol
- ▶ **Algorithm:** Dijkstra's Shortest Path First (SPF)
- ▶ **Metric:** Cost = 108 / Bandwidth (lower cost = preferred path)
- ▶ **Area:** Uses AREAS for scalability — Area 0 = Backbone Area
- ▶ **Updates:** Triggered only (**LSA flooding**) — **NOT periodic** like RIP
- ▶ **Convergence:** FAST convergence
- ▶ **Protocol:** Runs directly over IP (protocol number 89, **NOT TCP/UDP!**)

OSPF Router Types:

Router Type	Abbreviation	Role
Internal Router	IR	All interfaces in same area
Area Border Router	ABR	Connects multiple areas — connects to Area 0

Router Type	Abbreviation	Role
Autonomous System Boundary Router	ASBR	Connects OSPF to external routing protocols
Designated Router	DR	Elected per broadcast segment to reduce LSA flooding
Backup Designated Router	BDR	Backup for DR

OSPF Packet Types:

Type	Name	Purpose
Type 1	Hello	Discover/maintain neighbors, elect DR/BDR
Type 2	DBD (Database Description)	Summary of LSDB shared between neighbors
Type 3	LSR (Link State Request)	Request specific LSA from neighbor
Type 4	LSU (Link State Update)	Contains the actual LSA data
Type 5	LSAck (Link State Acknowledgment)	Confirms receipt of LSU

⚠ **EXAM TIP: OSPF is LINK STATE, uses Dijkstra, metric = COST. This is the most important routing protocol for exams!**

🟡 BGP — Border Gateway Protocol

- ▶ **Type:** Path Vector protocol (NOT distance vector or link state!)
- ▶ **Scope:** Inter-domain routing — used between different ISPs/ASes
- ▶ **Port:** TCP Port 179
- ▶ **Role:** The 'protocol of the Internet' — connects Autonomous Systems
 - eBGP = External BGP (between different AS)
 - iBGP = Internal BGP (within same AS)

▶ **Path Selection:** Based on policies, AS-path length, MED, LOCAL_PREF attributes

✓ **KEY POINT:** BGP is extremely stable but slow to converge — designed for policy-based routing

⚠ **EXAM TIP: BGP Hijacking = attacker announces false IP prefix routes to steal traffic. Major Internet security issue!**

SECTION 6: DATA LINK LAYER (Layer 2)

🔴 Ethernet — IEEE 802.3

- ▶ **Purpose:** Most widely used LAN technology
- ▶ **Frame Size:** Minimum 64 bytes, Maximum 1518 bytes (1500 bytes data + headers)
- ▶ **Addresses:** Uses MAC addresses (48-bit / 6 bytes, hardware address)
- ▶ **Speed:** 10 Mbps → 100 Mbps → 1 Gbps → 10 Gbps → 100 Gbps

⚡ CSMA/CD — Carrier Sense Multiple Access with Collision Detection

⚠ **EXAM TIP:** CSMA/CD is fundamental to Ethernet operation. Know all 5 steps!

Step	Action	Detail
1. Listen	Carrier Sense	Station listens to check if medium is free
2. Transmit	Multiple Access	If free, station transmits frame
3. Detect Collision	Collision Detection	If two stations transmit simultaneously — COLLISION
4. Jam Signal	Jam	Stations send jam signal to alert all of collision
5. Backoff	Random Backoff	Each station waits random time (Binary Exponential Backoff) before retry

✓ **KEY POINT:** CSMA/CA (Collision Avoidance) is used in WiFi (802.11) — NOT CSMA/CD!

✓ **KEY POINT:** CSMA/CD used in half-duplex Ethernet. Full-duplex Ethernet doesn't need it.

🟣 PPP — Point-to-Point Protocol

- ▶ **Use:** WAN connections — direct link between two nodes
- ▶ **OSI Layer:** Data Link Layer (Layer 2)
- ▶ **Features:** Encapsulation, authentication, compression, error detection
- ▶ **Authentication:** Supports PAP and CHAP

🔑 CHAP vs PAP Authentication:

Feature	PAP (Password Auth Protocol)	CHAP (Challenge Handshake Auth Protocol)
Security	WEAK — sends password in plaintext	STRONG — uses MD5 hash
Process	2-way handshake	3-way handshake (challenge-response)
Password sent?	YES — visible!	NO — only hash sent
Periodic reauth?	No	YES — periodic challenges

Feature	PAP (Password Auth Protocol)	CHAP (Challenge Handshake Auth Protocol)
Replay attack?	Vulnerable	Protected (random challenges)

⚠ **EXAM TIP:** CHAP is always preferred over PAP. PAP = insecure, sends password plaintext!

🟡 HDLC — High-Level Data Link Control

- ▶ **Type:** Bit-oriented synchronous data link layer protocol
- ▶ **Use:** WAN links, serial connections, predecessor to PPP
- ▶ **Standard:** ISO 13239 — based on IBM SDLC

📦 HDLC Frame Structure:

Field	Size	Purpose
Flag	8 bits	01111110 — Frame delimiter (marks start and end)
Address	8+ bits	Identifies secondary station
Control	8 or 16 bits	Frame type (I/S/U) and sequence numbers
Data (Information)	Variable	Actual payload data
FCS (Frame Check Sequence)	16 or 32 bits	CRC for error detection
Flag	8 bits	01111110 — End flag

- ▶ **Frame Types:** I-frames (Information), S-frames (Supervisory), U-frames (Unnumbered)

✓ **KEY POINT:** Bit stuffing: '0' inserted after 5 consecutive '1's to prevent flag emulation in data



SECTION 7: MASTER QUICK-REFERENCE TABLES



Complete Protocol Port Number Reference

Protocol	Port	Protocol	Port	Protocol	Port
FTP Data	20	SSH / SFTP	22	Telnet	23
SMTP	25	DNS	53	DHCP (Server)	67
DHCP (Client)	68	HTTP	80	POP3	110
NTP	123	IMAP	143	SNMP Agent	161
SNMP Trap	162	HTTPS	443	Kerberos	88
BGP	179	LDAP	389	LDAPS	636
FTP Control	21	RDP	3389	MySQL	3306

Security Attack Reference Table

Attack	Target Protocol/Layer	How It Works	Prevention
DNS Spoofing	DNS (App Layer)	Fake DNS response redirects user	DNSSEC, trusted resolvers
ARP Poisoning	ARP (Net Layer)	Fake ARP replies poison cache	Dynamic ARP Inspection
SYN Flood	TCP (Transport)	Half-open connections exhaust resources	SYN cookies, rate limiting
Man-in-Middle	Any layer	Intercepts communication	TLS/SSL, certificates
BGP Hijacking	BGP (Routing)	False route announcements	RPKI, route filtering
ICMP Smurf	ICMP (Network)	Amplified ping flood	Block broadcast pings
SSL Stripping	SSL/TLS	Downgrade HTTPS to HTTP	HSTS headers
Replay Attack	Auth protocols	Reuse captured auth data	Timestamps, nonces

OSI Layer Summary

Layer #	Layer Name	Key Protocols	PDU Name	Device
7	Application	HTTP, FTP, DNS, SMTP, SSH, Telnet	Data	Firewall, Proxy

Layer #	Layer Name	Key Protocols	PDU Name	Device
6	Presentation	SSL/TLS, JPEG, MPEG	Data	—
5	Session	NetBIOS, RPC, Kerberos	Data	—
4	Transport	TCP, UDP	Segment (TCP) / Datagram (UDP)	Load Balancer
3	Network	IP, ICMP, ARP, OSPF, BGP	Packet	Router
2	Data Link	Ethernet, PPP, HDLC, MAC	Frame	Switch, Bridge
1	Physical	Ethernet cables, Wi-Fi signal	Bit	Hub, Repeater



SECTION 8: MCQ TRICKS & MNEMONICS



Key MCQ Tricks — Memorize These!

MCQ Pattern	Answer	Remember With
Which protocol has NO encryption?	Telnet (Port 23)	T=Transparent (all visible!)
FTP uses which TWO ports?	20 (data) + 21 (control)	20=Data flows, 21=Commands
What does HTTPS add to HTTP?	TLS encryption	S = Secure = TLS
OSPF uses which algorithm?	Dijkstra's SPF	OSPF = Optimal Shortest Path First
RIP maximum hop count?	15 (16 = unreachable)	RIP = Really Insufficient for big nets
BGP runs on which port?	TCP 179	BGP=Big Gateway Protocol=179
Kerberos uses which encryption?	Symmetric	K=Key=Shared Secret
IPsec Tunnel Mode protects?	Entire IP packet	Tunnel = Everything inside
Which is faster: TCP or UDP?	UDP (no handshake)	UDP = Undoubtedly swift!
SMTP is for?	SENDING email only	SMTP=Send Mail To People
POP3 vs IMAP — which syncs?	IMAP (stays on server)	IMAP=I May Access Phones
ARP resolves?	IP → MAC	ARP=Address Resolution Protocol=IP→MAC



Protocol Mnemonics

Mnemonic	Remembers
Please Do Not Throw Sausage Pizza Away	OSI Layers 1-7: Physical, Data Link, Network, Transport, Session, Presentation, Application
DNS = Phone Book of Internet	DNS maps names to IP addresses
SSH = Secure SHell (replaces Telnet)	SSH = encrypted Telnet replacement
SYN → SYN-ACK → ACK	TCP 3-way handshake sequence
CSMA/CD = Listen → Talk → Crash → Backoff	Ethernet collision handling
ARP = Ask who has this IP? RARP = Ask who am I?	ARP vs RARP direction
BGP = Border of the Internet	BGP connects different ISPs/AS
IPsec Modes: Transport=Two hosts Tunnel=Two networks	IPsec mode selection rule



Last-Minute Flash Points

⚠ EXAM TIP: HTTP=80, HTTPS=443, SSH=22, Telnet=23, FTP=20/21, DNS=53, SMTP=25, POP3=110, IMAP=143

⚠ EXAM TIP: TCP = Reliable + Connection-oriented | UDP = Unreliable + Connectionless

⚠ EXAM TIP: OSPF = Link State, Dijkstra, Cost metric | RIP = Distance Vector, Bellman-Ford, Hop metric

⚠ EXAM TIP: IPsec Transport = host-to-host (payload only) | Tunnel = network-to-network (full packet)

⚠ EXAM TIP: TLS = current standard (SSL deprecated) | ARP Poisoning = major network attack

⚠ EXAM TIP: Kerberos = ticket-based, symmetric encryption | PGP = email encryption, web of trust

⚠ EXAM TIP: CHAP > PAP (CHAP uses challenge-response, PAP sends plaintext password)

⚠ EXAM TIP: BGP = Path Vector, TCP port 179 = Inter-domain Internet routing backbone

BEST OF LUCK FOR YOUR EXAM!

Information Security Scientist B — Computer Science